CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

ITUE203 – Web Development Frameworks

PRACTICAL 2

Semantic HTML5 Pages with Accessibility-Ready Structure

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Semester / Academic Year	3 / 2026-27
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1. Aim and Objectives

To develop static HTML5 skeletons for the StudentHub portal pages using semantic elements and an accessibility-ready page structure.

- Build at least 10 linked static pages for StudentHub. This practical provides 17: the 11 pages named in the problem definition and 6 additional student-area pages.
- Use semantic HTML5 elements – header, nav, main, section, article, aside and footer – to describe the meaning of each part of a page.
- Make every page accessible: one h1 with ordered sub-headings, labelled form controls, descriptive image alt text, a skip-to-content link, a breadcrumb and aria-current on the current page.
- Keep the structure identical on all pages so that navigation is predictable.
- Record the browser output of every page and check the structure with a repeatable script.

2. Problem Definition

Official problem definition: *"Develop static HTML5 skeletons for at least 10 StudentHub pages such as Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback. Use semantic tags and accessibility-friendly structure."*

A student portal is used every day by many students, including people who navigate with a keyboard or a screen reader. If every page is built differently, users have to relearn the layout on each page, and assistive technology cannot tell the menu from the content. Designing the pages first as clean semantic skeletons gives StudentHub a consistent structure – the same header, navigation, main area and footer everywhere – on which styling (Practical 3) and interactivity can be added later without rewriting the markup.

The practical list asks three key questions; the answers for this implementation are:

Key question	Answer in this practical
Are semantic tags such as header, nav, main, section, article, aside and footer used properly?	Yes. Every page has one header, a main navigation and a breadcrumb (both nav), exactly one main and a footer. Content is grouped in section, article and aside elements (Section 4).
Are form labels, headings, image alt text and navigation links accessible?	Every form control has a label, every page has one h1 with no skipped levels, the logo has descriptive alt text, and the current page is marked with aria-current (Sections 4 and 7).
Is the page structure consistent across all pages?	Yes. All 17 pages share the same skeleton and the same main navigation; this was checked by the local script in Section 7.

3. Requirements and Tools

3.1 Requirements

- **Pages:** at least 10 static pages, including Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ and Feedback.
- **Semantics:** semantic HTML5 elements instead of generic containers; correct heading hierarchy.
- **Accessibility:** labelled forms, image alt text, accessible navigation links.
- **Linking:** every page reachable from every other page through local links.

- **Extensions from the practical list:** breadcrumb navigation (intermediate) and a skip-to-content link with keyboard-friendly navigation (advanced). Both are implemented.
- **Scope:** HTML only – no CSS, JavaScript or server code in this practical.

3.2 Tools

Tool	Use in this practical
HTML5	Markup of all 17 pages
Visual Studio Code	Source code editor; used for the VS Code source screenshots (Section 5.6)
Microsoft Edge 156.0.4314.50	Viewing the pages; headless mode used to capture the browser screenshots in Section 6
Python 3.12.0	Runs Practical-2/source/check_pages.py, the local structure, link and accessibility checker (standard library only)
Git	Version control of the StudentHub repository
Operating system	Windows 11

4. Page Structure and Implementation

4.1 Common page skeleton

Every page is built from the same skeleton, in this order:

Element	Purpose
<a href="#main-content">	Skip-to-content link. It is the first link on the page, so a keyboard user can jump past the header and menu with one Tab and Enter.
<header>	Site banner: the StudentHub logo (a link to Home) and the site name.
<nav aria-label="Main navigation">	Links to all 17 pages in three groups: Public pages, Student area and Administration. The current page is marked with aria-current="page" and shown in bold.
<nav aria-label="Breadcrumb">	Shows the position of the page, e.g. Home > Dashboard > Attendance. The last item carries aria-current="page"; the > separators are hidden from screen readers with aria-hidden.
<main id="main-content">	The unique main content of the page, starting with its single h1. Content is grouped in section, article and aside elements with h2 and h3 headings.
<footer>	Links to Contact, FAQ and Feedback, and the copyright line.

4.2 Accessibility features

- **Headings:** each page has exactly one h1 (the page title); h2 starts each section and h3 is used inside sections and articles. No level is skipped.
- **Forms and labels:** every input, select and textarea has a label linked with for/id or wraps the control. Related fields are grouped with fieldset and legend; radio groups (gender, rating) have their own legend. autocomplete values (name, email, username, current-password, new-password) are set on login, register and profile fields.

- **Honest static forms:** Practical 2 has no server, so every submit button is disabled and each form states that nothing is sent or saved. The Clear (reset) buttons and the details/summary sections work without JavaScript.
- **Images:** the only image is the StudentHub logo, with the alt text “StudentHub logo – go to the Home page”, because the image is a link.
- **Navigation:** the two nav elements have different aria-label values so screen readers can tell them apart; the current page has aria-current="page".
- **Tables:** data tables have a caption and th headers with scope="col" or scope="row". border="1" keeps them readable without CSS (an obsolete but permitted HTML attribute).
- **Keyboard:** only native links, buttons, form controls and details elements are used; there is no tabindex and no click-only element.
- **Language and metadata:** lang="en", UTF-8 charset, a viewport meta tag, a description and a unique title on every page.

4.3 Required pages

#	Page	File	Content and main elements
1	Home	index.html	Welcome text; three article blocks for portal sections; guidelines; Quick FAQ with details/summary; aside with sample campus alerts
2	About	about.html	Purpose of StudentHub; user roles in a description list (dl); list of modules linking every page; note about this practical
3	Register	register.html	Registration form with three fieldsets (personal, academic, account), a gender radio group with legend and a terms checkbox
4	Login	login.html	Sign-in form (role, enrollment number or email, password); “Forgot your password?” in details/summary; link to Register
5	Dashboard	dashboard.html	Academic summary table with meter and progress elements; weekly timetable table; aside with an event alert
6	Events	events.html	Two event articles with time elements for the dates; how events are published
7	Profile	profile.html	Personal details in a dl; contact-details form; change-password form inside details/summary
8	Contact	contact.html	Department address in an address element; inquiry form
9	Admin	admin.html	In-page navigation to five sections; overview (dl); tables of students, events, users and an audit log; add-student and publish-event forms
10	FAQ	faq.html	Four sections of questions, each answer in details/summary
11	Feedback	feedback.html	Feedback form: page select, rating radio group with legend, comments

4.4 Additional pages

#	Page	File	Content and main elements
12	Courses	courses.html	Subjects table with a tfoot total; syllabus outline in details/summary
13	Attendance	attendance.html	Subject-wise attendance table with meter elements; leave application form
14	Assignments	assignments.html	Assignment status table; submission form with a file input
15	Results	results.html	Semester-wise SGPA/CGPA table; Semester 2 grade summary table
16	Notices	notifications.html	Three notice articles with posting dates
17	Directory	directory.html	Search form; student directory table

4.5 Content and limitations

- Content marked “Sample data” is for demonstration only. Figures such as attendance, CGPA and the timetable are placeholders, not actual academic records.
- The Admin and Directory pages contain only fictional entries (for example “Sample Student 1”, “DEMO-001”, “sample.admin”). No real student data, accounts or passwords appear in the pages.
- The Admin Console is reachable from the menu because a static page cannot check a login; access control belongs to the server-side application.
- The pages are intentionally unstyled. Layout and visual design are the subject of Practical 3.

5. Source Code Evidence

The excerpts below are copied unchanged from the files in Practical-2/. Line numbers match the files; only the common leading indentation of each excerpt is removed.

File	Lines	Shows
Practical-2/index.html	1–19	Document head, skip link, header and the start of the main navigation
Practical-2/index.html	47–64	Breadcrumb, main element, h1, section and article
Practical-2/index.html	98–112	aside and footer
Practical-2/register.html	59–79, 116–127	Form with fieldset, legend and labels; disabled submit button
Practical-2/admin.html	53–66, 80–98	In-page navigation, section and an accessible data table

5.1 index.html – head, skip link and header

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="StudentHub home page: student portal for CSE students of CSPIT, CHARUSAT.">
    <title>Home – StudentHub</title>
</head>
<body>
    <a href="#main-content">Skip to main content</a>

    <header>
        <a href="index.html"><img src="images/studenthub-logo.svg" alt="StudentHub logo – go to the Home page" width="48" height="48"></a>
        <p><strong>StudentHub</strong> – Student portal for Computer Science &amp;amp; Engineering, CSPIT, CHARUSAT</p>
        <nav aria-label="Main navigation">
            <ul>
                <li>Public pages
                    <ul>
                        <li><a href="index.html" aria-current="page"><strong>Home</strong></a></li>
```

- **Document setup:** <!DOCTYPE html>, lang="en", UTF-8, viewport, description and a page-specific title.
- **Skip link:** the first element in the body, pointing to #main-content.
- **Header and navigation:** the logo link has descriptive alt text and fixed width and height; the main nav is a nested list, and the current page has aria-current="page".

5.2 index.html – breadcrumb, main and articles

```
    </nav>
    <nav aria-label="Breadcrumb">
        <p>You are here: <span aria-current="page">Home</span></p>
    </nav>
</header>

<main id="main-content">
    <h1>Welcome to StudentHub</h1>
    <p>Your portal for Computer Science &amp;amp; Engineering (CSE) at CSPIT, A7 Building, CHARUSAT. Access your timetable, Semester 3 subjects, attendance records and exam results.</p>
    <p><a href="login.html">Login</a> | <a href="register.html">Register as a new student</a></p>

    <section>
        <h2>Portal sections</h2>
        <article>
            <h3>Semester 3 subjects and timetable</h3>
            <p>Check your enrolled CSE subjects, including FDSA, WDF, OOP, FCN and Discrete Mathematics.</p>
            <p><a href="courses.html">View courses</a> | <a href="dashboard.html">View timetable on the dashboard</a></p>
        </article>
```

- **Breadcrumb:** a second nav with its own aria-label; on the Home page it contains only the current page.
- **Main content:** main has id="main-content" (the skip-link target) and starts with the page's only h1.
- **Grouping:** a section with an h2 contains independent article blocks, each with an h3.

5.3 index.html – aside and footer

```
    <aside aria-labelledby="alerts-heading">
        <h2 id="alerts-heading">Latest campus alerts</h2>
        <p><mark>Sample data:</mark> example notices for this static page.</p>
        <ul>
            <li><time datetime="2026-08-22">22 August 2026</time>: Odoo Hackathon at L.D. College of Engineering.</li>
            <li><time datetime="2026-08-15">15 August 2026</time>: Independence Day celebration at CHARUSAT Campus.</li>
        </ul>
        <p><a href="notifications.html">See all notices</a></p>
    </aside>
</main>

<footer>
    <p><a href="contact.html">Contact</a> | <a href="faq.html">FAQ</a> | <a href="feedback.html">Feedback</a></p>
    <p>&copy; <time datetime="2026">2026</time> StudentHub – ITUE203 Practical 2, static HTML5 skeleton by Shreyas (25CS023). Content marked “Sample data” is for demonstration only.</p>
</footer>
```

- **Complementary content:** the aside is labelled by its own h2 through aria-labelledby; dates use the time element with a machine-readable datetime.
- **Footer:** repeated on every page with links to Contact, FAQ and Feedback.

5.4 register.html – accessible form

```
<form>
    <fieldset>
        <legend>Personal details</legend>
        <p>
            <label for="reg-name">Full name *</label><br>
            <input type="text" id="reg-name" name="name" autocomplete="name" required>
        </p>
        <p>
            <label for="reg-email">College email address *</label><br>
            <input type="email" id="reg-email" name="email" autocomplete="email" required>
        </p>
        <p>
            <label for="reg-mobile">Mobile number *</label><br>
            <input type="tel" id="reg-mobile" name="mobile" autocomplete="tel" maxlength="10" required>
        </p>
        <fieldset>
            <legend>Gender *</legend>
            <label><input type="radio" name="gender" value="male" required> Male</label>
            <label><input type="radio" name="gender" value="female"> Female</label>
            <label><input type="radio" name="gender" value="other"> Other</label>
        </fieldset>
```

```
        <p>
            <input type="checkbox" id="reg-terms" name="terms" required>
            <label for="reg-terms">I agree to the terms and conditions and the privacy policy. *</label>
        </p>
    </fieldset>

    <p><strong>Note:</strong> This is a static form skeleton. Practical 2 has no server, so the “Register account” button is disabled and nothing is sent or saved.</p>
    <p>
        <button type="submit" disabled>Register account</button>
        <button type="reset">Clear</button>
    </p>
</form>
```

- **Labels:** each label’s for attribute matches the id of its input; the radio buttons are wrapped in their labels.

- **Grouping:** fieldset and legend group related fields; the gender radio group is a nested fieldset with its own legend.
- **Input types and hints:** type="email", type="tel" and type="password", required, maxlength, minlength and autocomplete help browsers and assistive technology.
- **No pretend backend:** the submit button is disabled and the note above it says that nothing is sent or saved. The form has no action attribute.

5.5 admin.html – in-page navigation and data table

```
53  <main id="main-content">
54      <h1>Admin Console</h1>
55      <p><mark>Sample data:</mark> every record on this page is a fictional demonstration entry. It
    contains no real student data, accounts or passwords.</p>
56      <p>In the full application only administrators can open this page. This static skeleton has no
    login check.</p>
57
58      <nav aria-label="Admin sections">
59          <ul>
60              <li><a href="#overview">Overview</a></li>
61              <li><a href="#students">Students</a></li>
62              <li><a href="#events">Events</a></li>
63              <li><a href="#users">Users</a></li>
64              <li><a href="#audit-log">Audit log</a></li>
65          </ul>
66      </nav>
```

```
80  <section id="students">
81      <h2>Students</h2>
82      <table border="1">
83          <caption>Student records (sample entries)</caption>
84          <thead>
85              <tr>
86                  <th scope="col">Enrollment</th>
87                  <th scope="col">Name</th>
88                  <th scope="col">Course</th>
89                  <th scope="col">Year</th>
90              </tr>
91          </thead>
92          <tbody>
93              <tr>
94                  <th scope="row">DEMO-001</th>
95                  <td>Sample Student 1</td>
96                  <td>B.Tech CSE</td>
97                  <td>2nd year</td>
98              </tr>
```

- **Sample data notice:** the page states at the top that every record is fictional.
- **In-page navigation:** a third nav, labelled “Admin sections”, links to the ids of the five sections.
- **Accessible table:** the caption names the table; column headers use scope="col" and the first cell of each row is a th with scope="row".

5.6 Screenshots from VS Code

PLACEHOLDER – VS Code screenshot: Practical-2/index.html (head, skip link, header, navigation)

Insert a genuine screenshot of the file open in VS Code.

PLACEHOLDER – VS Code screenshot: Practical-2/register.html (form, fieldset, legend, labels)

Insert a genuine screenshot of the file open in VS Code.

PLACEHOLDER – VS Code screenshot: Practical-2/admin.html (sections and table)

Insert a genuine screenshot of the file open in VS Code.

6. Browser Output

The screenshots below are the actual pages rendered by Microsoft Edge 156 in headless mode at a width of 1280 pixels. Each shows the full page from the skip link to the footer; only the blank space below the footer was trimmed. The pages are unstyled because Practical 2 uses no CSS. The original image files are in Practical-2/screenshots/browser/.

6.1 Home – index.html

Skip to main content

StudentHub – Student portal for Computer Science & Engineering, CSPIT, CHARUSAT

- Public pages
  - **Home**
  - About
  - Events
  - FAQ
  - Contact
  - Feedback
  - Login
  - Register
- Student area
  - Dashboard
  - My Profile
  - Courses
  - Attendance
  - Assignments
  - Results
  - Notices
  - Directory
- Administration
  - Admin Console

You are here: Home

Welcome to StudentHub

Your portal for Computer Science & Engineering (CSE) at CSPIT, A7 Building, CHARUSAT. Access your timetable, Semester 3 subjects, attendance records and exam results.

Login | Register as a new student

Portal sections

Semester 3 subjects and timetable

Check your enrolled CSE subjects, including FDSA, WDF, OOP, FCN and Discrete Mathematics.

View courses | View timetable on the dashboard

Attendance and results

Monitor your subject-wise attendance and check the results of previous semesters.

Check attendance | View results

Assignments, events and notices

Semester 3 assignments are coming soon. Stay updated with campus hackathons and activities.

View assignments | View events | Read notices

Portal guidelines

All assignments must be submitted on time. For technical issues with the portal, use the Contact page.

Quick FAQ

▶ Where is the CSE Department located?
▶ How do I check my Semester 3 schedule?
▶ Who do I contact for portal queries?

Latest campus alerts

Sample data: example notices for this static page.

- 22 August 2026: Odoo Hackathon at L.D. College of Engineering.
- 15 August 2026: Independence Day celebration at CHARUSAT Campus.

See all notices

Contact | FAQ | Feedback

© 2026 StudentHub – ITUE203 Practical 2, static HTML5 skeleton by Shreyas (25CS023). Content marked "Sample data" is for demonstration only.

Figure 1: Home page (index.html) in Microsoft Edge

6.2 About – about.html

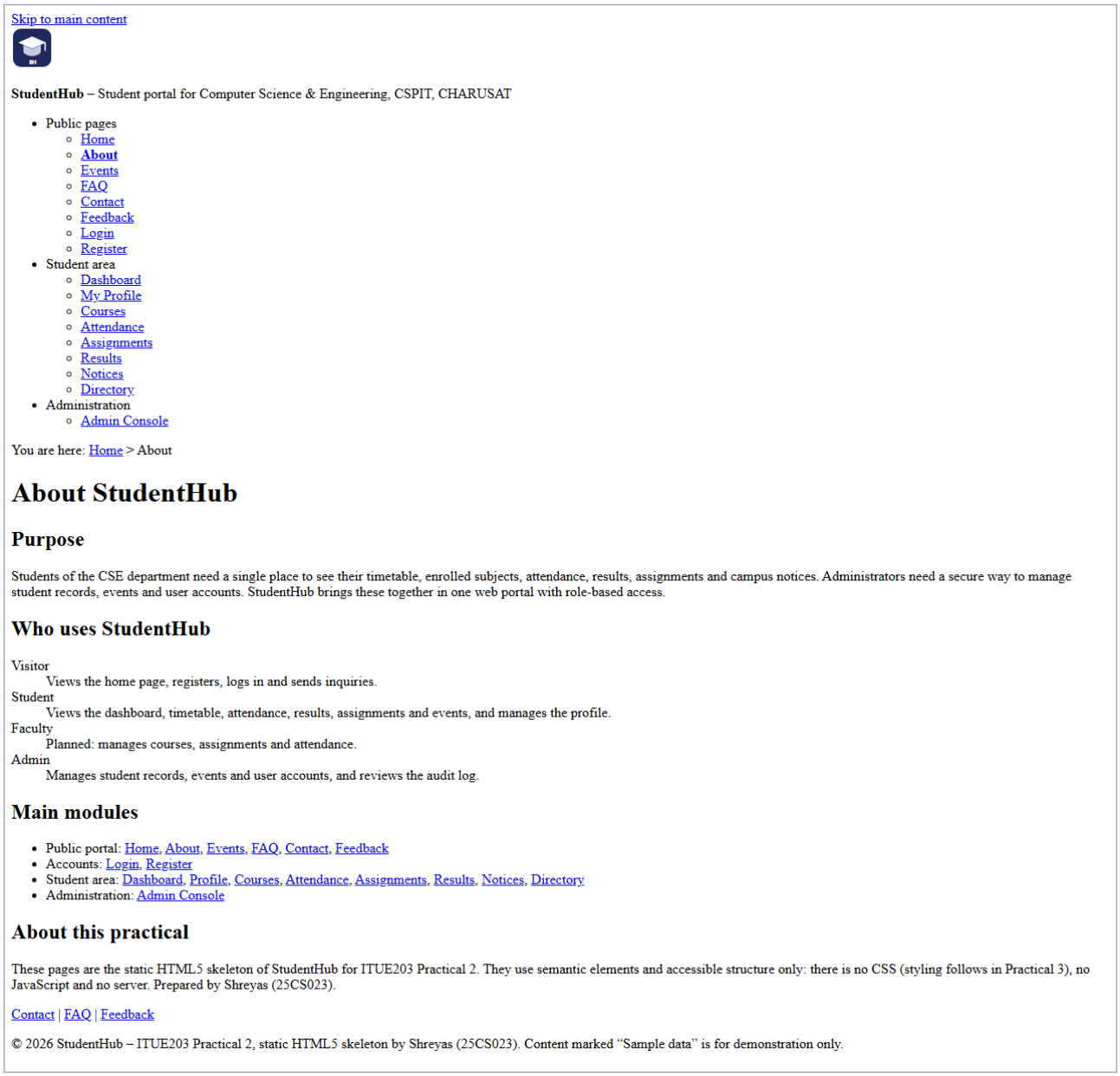


Figure 2: About page (about.html) in Microsoft Edge

6.3 Register – register.html

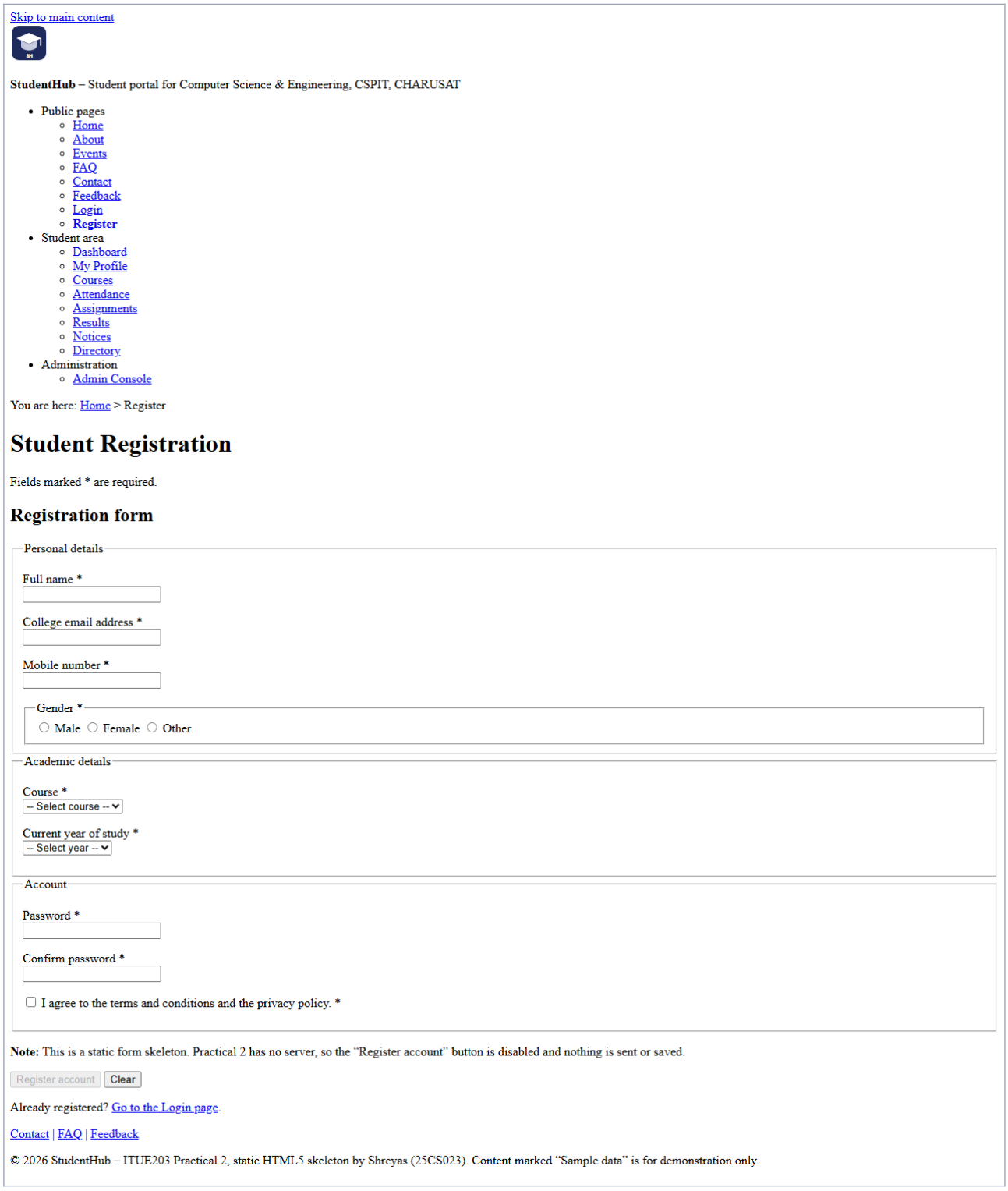


Figure 3: Register page (register.html) in Microsoft Edge

6.4 Login – login.html

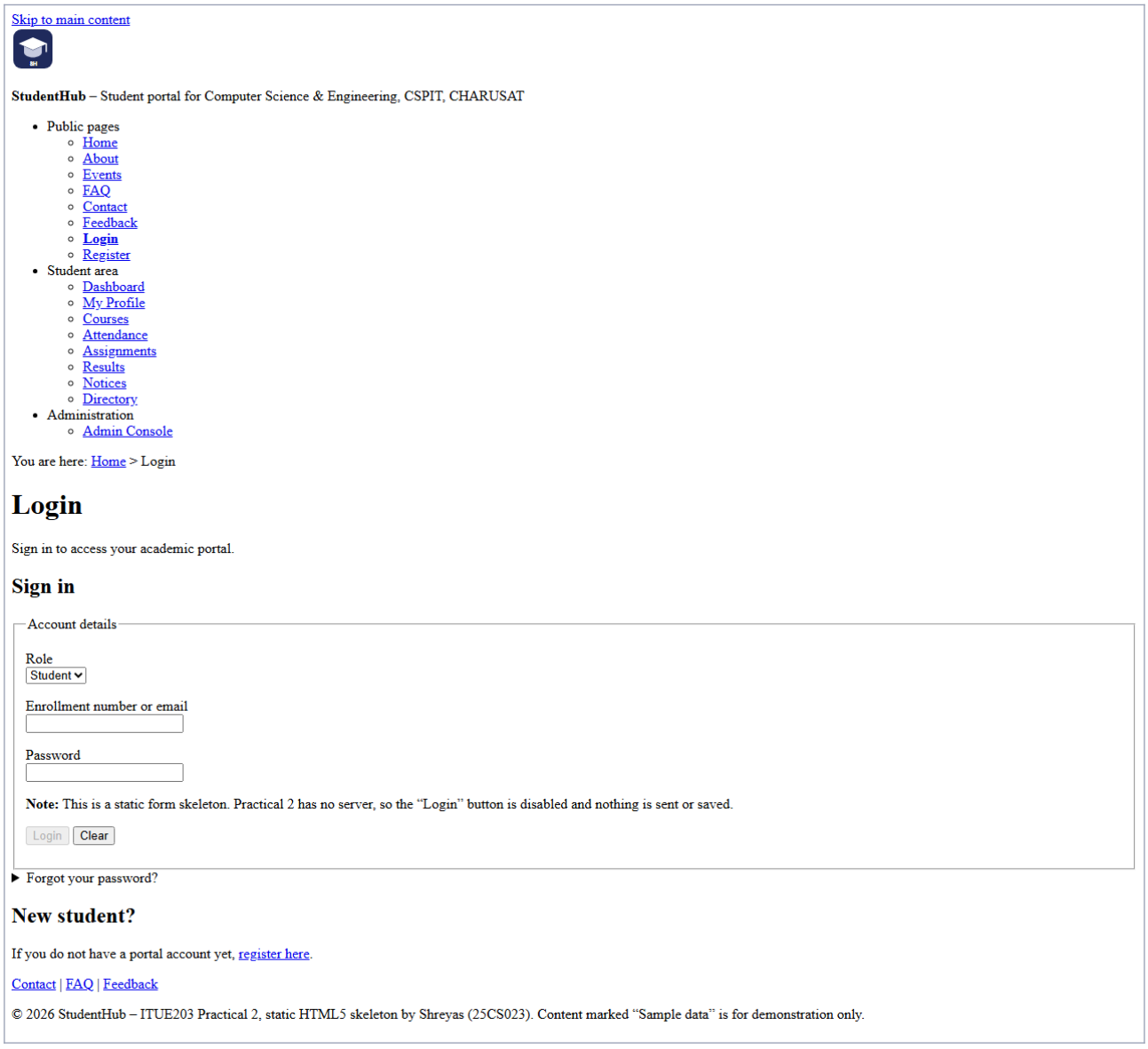


Figure 4: Login page (login.html) in Microsoft Edge

6.5 Dashboard – dashboard.html

Skip to main content

StudentHub – Student portal for Computer Science & Engineering, CSPIT, CHARUSAT

- Public pages
  - Home
  - About
  - Events
  - FAQ
  - Contact
  - Feedback
  - Login
  - Register
- Student area
  - **Dashboard**
  - My Profile
  - Courses
  - Attendance
  - Assignments
  - Results
  - Notices
  - Directory
- Administration
  - Admin Console

You are here: Home > Dashboard

Student Dashboard

Welcome, **Shreyas** | Enrollment: 25CS023 | B.Tech CSE, Semester 3, CSPIT

Sample data: the figures and the timetable on this page are placeholders, not actual academic records.

Semester 3 academic summary

Academic summary for Semester 3

Category	Current status	Progress	Details
Overall attendance	94.0%		View attendance
Assignments	Coming soon for Semester 3		View assignments
Overall CGPA	8.05 / 10.0		View results
Enrolled credits	21 credits (6 subjects)		View courses

Weekly timetable

Semester 3 weekly timetable, Room 506, A7 Building

Day	09:10–10:10	10:10–11:10	11:10–12:10	12:10–14:10	14:20–16:20
Monday	FDSA (VMP)	FCN (HYY)	Lunch break	HS (ADK)	Labs: A1-FDSA / B1-FCN / C1-WDF
Tuesday	MATHS (PM)	FDSA (ART)	Lunch break	Labs: A1-FCN / B1-WDF / C1-FCN	FCN (HYY) and WDF (SRG)
Wednesday	MATHS (PM)	OOP (SSS)	Lunch break	Labs: A1-WDF / B1-WDF / C1-FDSA	WDF (SRG) and FDSA (VMP)
Thursday	WDF (VYP)	FCN (HYY)	Lunch break	Labs: A1-WDF / B1-FDSA / C1-WDF	Labs: A1-OOP / B1-WDF / C1-OOP
Friday	MATHS (PM)	FDSA (ART)	Lunch break	Counselling sessions (Rooms 631–633)	FCN (HYY) and OOP (SSS)

Event alert

Odoo Hackathon at LDCE

Date: 22 August 2026

A hackathon at L.D. College of Engineering. CSE students can register as teams.

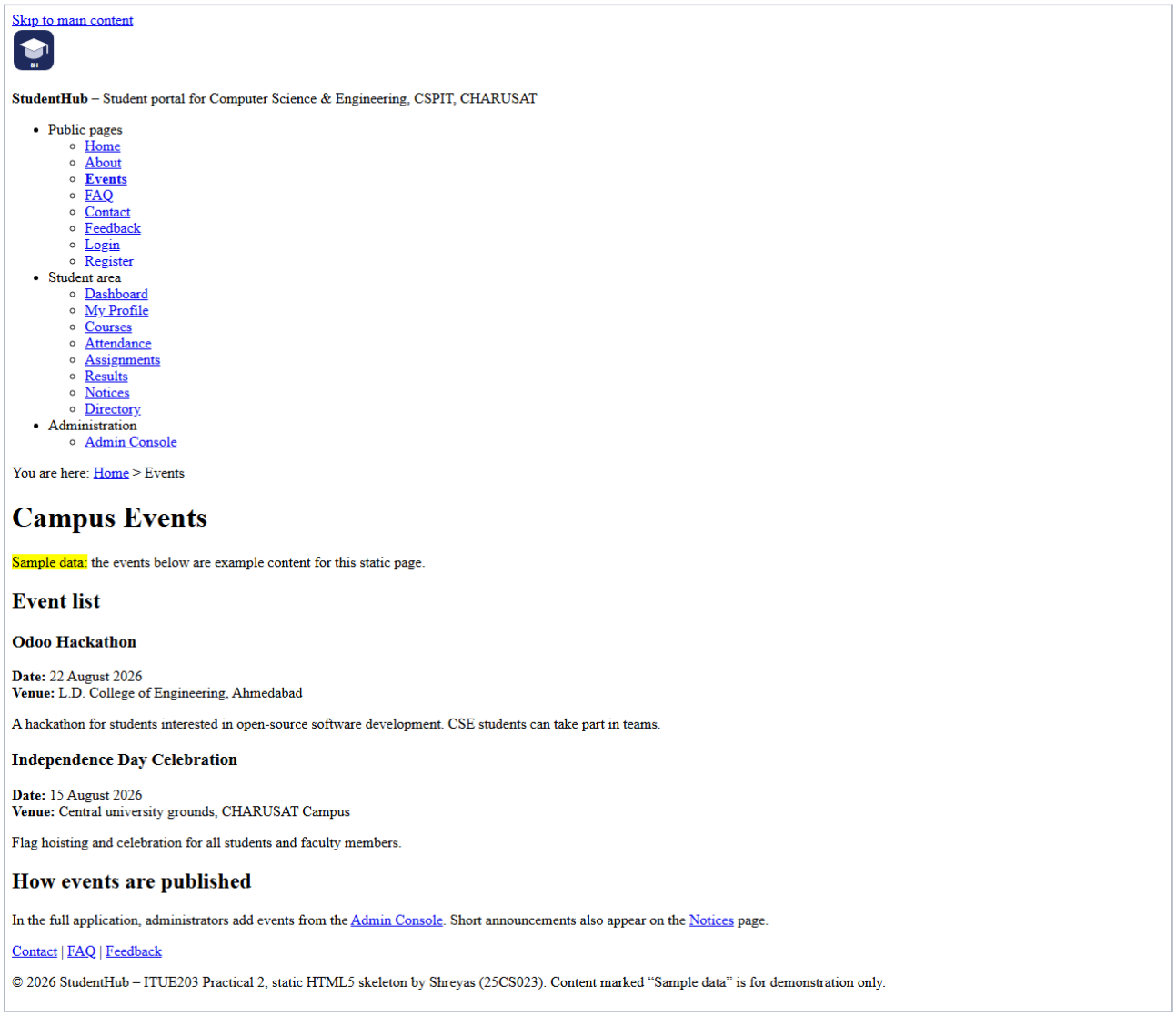
See all events | See all notices

Contact | FAQ | Feedback

© 2026 StudentHub – ITUE203 Practical 2, static HTML5 skeleton by Shreyas (25CS023). Content marked "Sample data" is for demonstration only.

Figure 5: Dashboard page (dashboard.html) in Microsoft Edge

6.6 Events – events.html



Skip to main content

StudentHub – Student portal for Computer Science & Engineering, CSPIT, CHARUSAT

- Public pages
  - Home
  - About
  - **Events**
  - FAQ
  - Contact
  - Feedback
  - Login
  - Register
- Student area
  - Dashboard
  - My Profile
  - Courses
  - Attendance
  - Assignments
  - Results
  - Notices
  - Directory
- Administration
  - Admin Console

You are here: Home > Events

Campus Events

Sample data: the events below are example content for this static page.

Event list

Odoo Hackathon

Date: 22 August 2026
Venue: L.D. College of Engineering, Ahmedabad

A hackathon for students interested in open-source software development. CSE students can take part in teams.

Independence Day Celebration

Date: 15 August 2026
Venue: Central university grounds, CHARUSAT Campus

Flag hoisting and celebration for all students and faculty members.

How events are published

In the full application, administrators add events from the Admin Console. Short announcements also appear on the Notices page.

Contact | FAQ | Feedback

© 2026 StudentHub – ITUE203 Practical 2, static HTML5 skeleton by Shreyas (25CS023). Content marked "Sample data" is for demonstration only.

Figure 6: Events page (events.html) in Microsoft Edge

6.7 Profile – profile.html

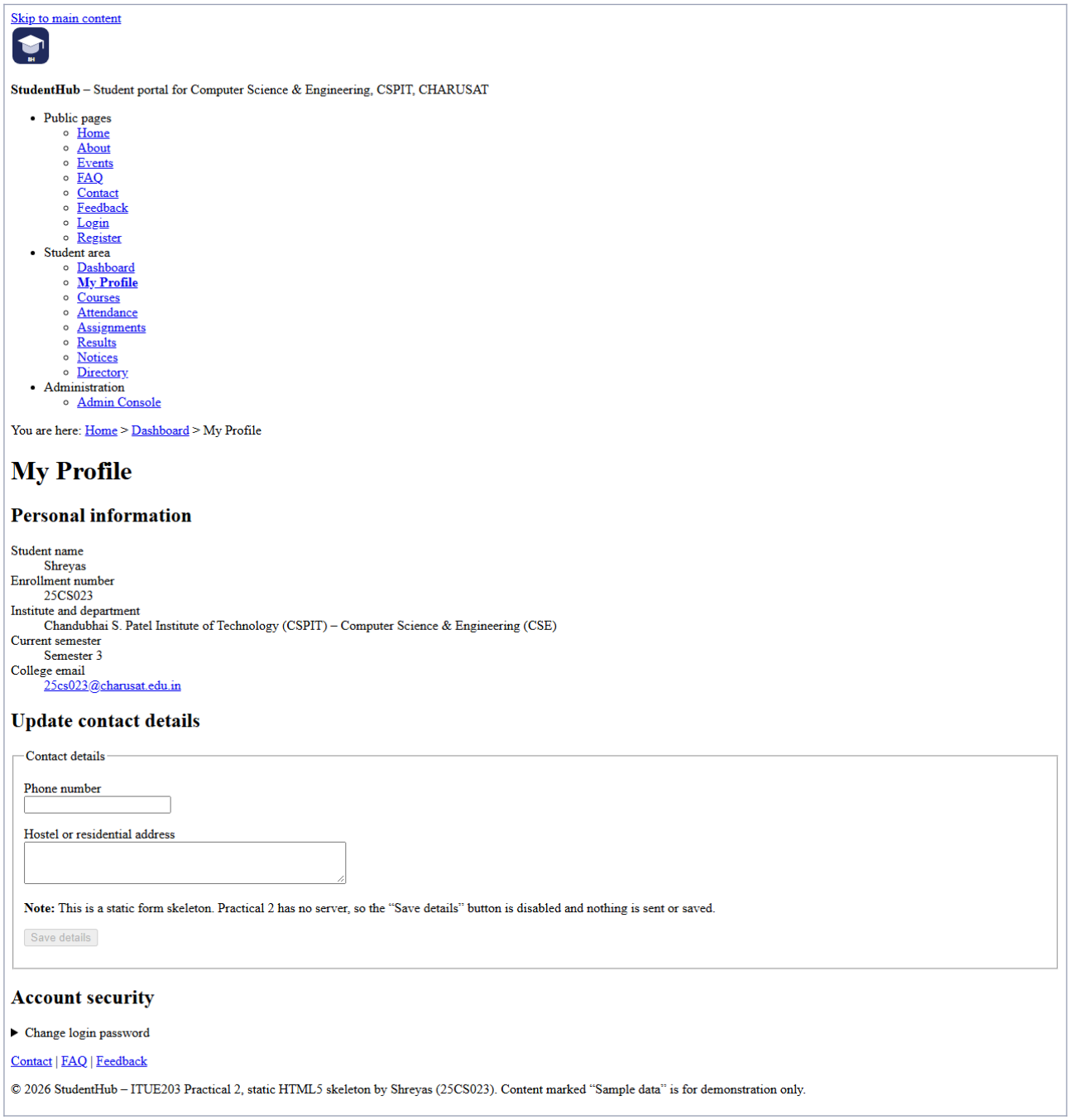


Figure 7: Profile page (profile.html) in Microsoft Edge

6.8 Contact – contact.html



Figure 8: Contact page (contact.html) in Microsoft Edge

6.9 Admin – admin.html

The Admin Console is longer than one page, so its screenshot (09-admin.png) is shown in two parts, split at a blank line between the Students and Events sections.

Skip to main content

StudentHub – Student portal for Computer Science & Engineering, CSPIT, CHARUSAT

- Public pages
  - Home
  - About
  - Events
  - FAQ
  - Contact
  - Feedback
  - Login
  - Register
- Student area
  - Dashboard
  - My Profile
  - Courses
  - Attendance
  - Assignments
  - Results
  - Notices
  - Directory
- Administration
  - **Admin Console**

You are here: Home > Admin Console

Admin Console

Sample data: every record on this page is a fictional demonstration entry. It contains no real student data, accounts or passwords.

In the full application only administrators can open this page. This static skeleton has no login check.

- Overview
- Students
- Events
- Users
- Audit log

Overview

Student records
5
Published events
2
User accounts
3

Students

Student records (sample entries)

Enrollment	Name	Course	Year
DEMO-001	Sample Student 1	B.Tech CSE	2nd year
DEMO-002	Sample Student 2	B.Tech CSE	2nd year
DEMO-003	Sample Student 3	B.Tech IT	2nd year
DEMO-004	Sample Student 4	B.Tech CE	3rd year
DEMO-005	Sample Student 5	BCA	1st year

Add a student record

Figure 9a: Admin page (admin.html), part 1 of 2 – header, overview and students

New student

Full name

Enrollment number

Course

-- Select course --

Note: This is a static form skeleton. Practical 2 has no server, so the "Add student" button is disabled and nothing is sent or saved.

Add student

Events

Published events (sample entries)

Title	Date	Venue
Odoo Hackathon	22 August 2026	L.D. College of Engineering
Independence Day Celebration	15 August 2026	CHARUSAT Campus

Publish an event

New event

Event title

Event date

dd-mm-yyyy

Poster image (JPG or PNG)

Choose File No file chosen

Note: This is a static form skeleton. Practical 2 has no server, so the "Publish event" button is disabled and nothing is sent or saved.

Publish event

Users

User accounts (sample entries, no passwords shown or stored)

Username	Role	Status
sample.admin	Admin	Active
sample.student1	Student	Active
sample.student2	Student	Active

Audit log

Recent account activity (sample entries)

Time	User	Action	Result
1 October 2026, 09:15	sample.admin	Login	Success
1 October 2026, 09:20	sample.admin	Published event	Success
1 October 2026, 10:02	sample.student1	Login	Failed (wrong password)

Contact | FAQ | Feedback

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Figure 9b: Admin page (admin.html), part 2 of 2 – events, users and audit log

6.10 FAQ – faq.html

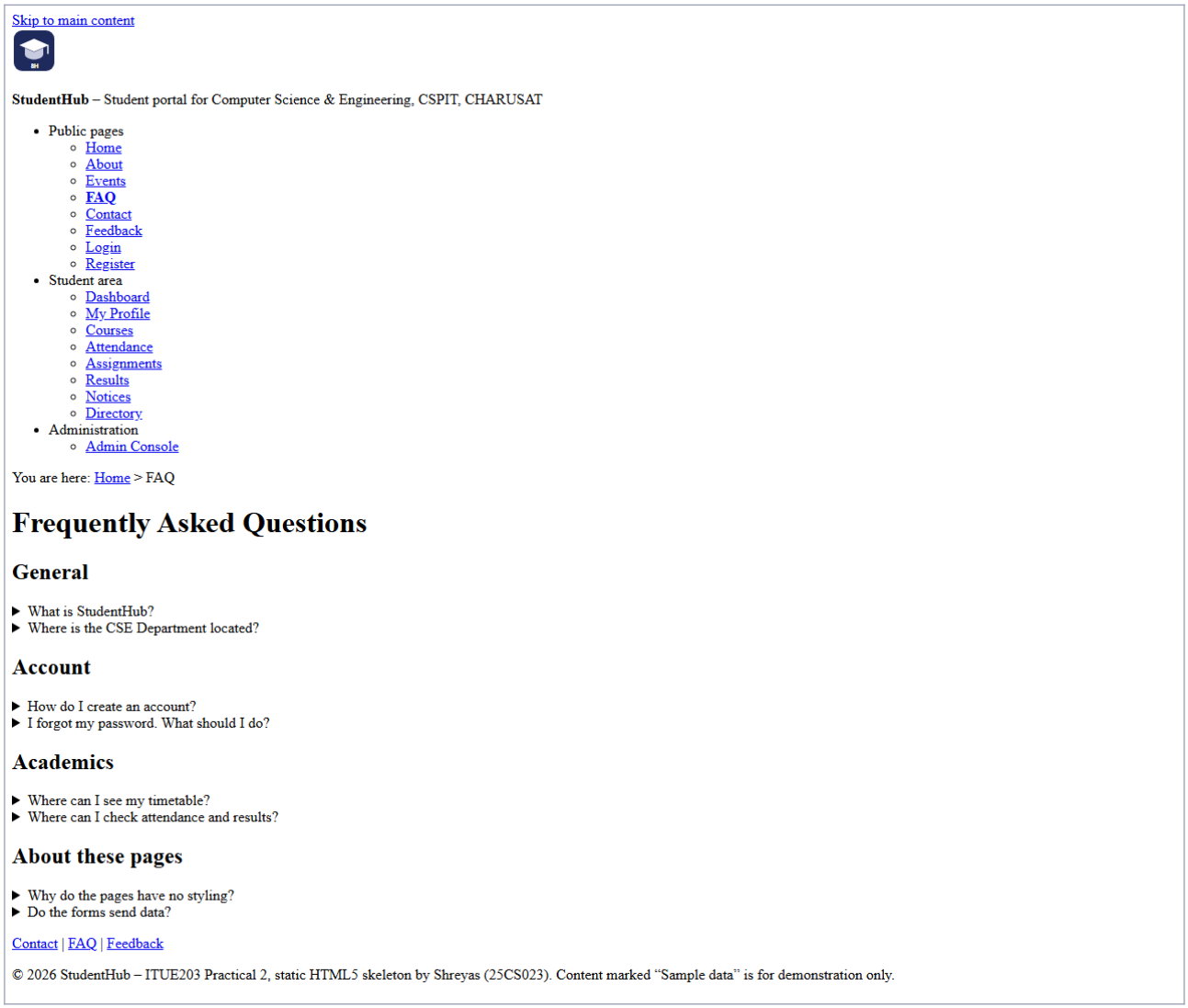


Figure 10: FAQ page (faq.html) in Microsoft Edge

6.11 Feedback – feedback.html



Figure 11: Feedback page (feedback.html) in Microsoft Edge

7. Testing and Accessibility Checklist

7.1 Local checker: source/check_pages.py

The pages were checked with Practical-2/source/check_pages.py, a script written for this practical using only the Python standard library. It is a local checker, not the official W3C Markup Validation Service: it tests the rules listed below, tag nesting and duplicate ids, but not every rule of the HTML specification.

Command used:

```
python Practical-2/source/check_pages.py Practical-2
```

Result: all 17 pages passed and the script printed "ALL CHECKS PASSED". The counts below are taken from the script's output; "links checked" counts every link on the page, including the shared navigation:

Page	Links checked	Form controls	Result
index.html	36	0	Passed
about.html	40	0	Passed
events.html	25	0	Passed
faq.html	29	0	Passed
contact.html	23	4	Passed
feedback.html	23	7	Passed
login.html	25	3	Passed
register.html	24	11	Passed
dashboard.html	29	0	Passed
profile.html	25	4	Passed
courses.html	24	0	Passed
attendance.html	24	3	Passed
assignments.html	24	2	Passed
results.html	24	0	Passed
notifications.html	26	0	Passed
directory.html	25	2	Passed
admin.html	28	6	Passed

As a control test, the same script was run on the earlier version of the Practical-2 folder (before this practical's changes). It reported 212 problems across the 17 expected files, including the five pages that did not exist yet; other examples were missing main navigation and skip links, table headers without scope, and obsolete table attributes. This shows that the script does detect the faults it is meant to find.

7.2 Accessibility checklist

#	Check	Result	How it was checked
1	DOCTYPE, lang="en", UTF-8 charset, viewport meta and a unique title on every page	Pass	check_pages.py
2	Semantic landmarks: header, nav, exactly one main and footer; content in section, article or aside	Pass	check_pages.py
3	Exactly one h1 per page; no skipped heading levels	Pass	check_pages.py
4	Skip-to-content link is the first link and targets main#main-content	Pass	check_pages.py, screenshots
5	Main navigation lists all 17 pages; current page marked with aria-current="page"	Pass	check_pages.py, screenshots
6	Breadcrumb navigation with aria-current="page"	Pass	check_pages.py, screenshots
7	Every form control has a label; radio groups use fieldset and legend	Pass	check_pages.py

#	Check	Result	How it was checked
8	Every image has descriptive alt text	Pass	check_pages.py
9	Data tables have a caption and th elements with scope	Pass	check_pages.py
10	All local and in-page links resolve; no href="#" placeholders; no external links	Pass	check_pages.py
11	Keyboard friendly: native elements only, no tabindex, no click-only controls	Pass	Source review
12	No non-working controls: submit buttons disabled with an explanation	Pass	check_pages.py, screenshots
13	No inline styles, style or script elements, or obsolete layout attributes	Pass	check_pages.py
14	Pages render correctly in a browser	Pass	All 17 screenshots opened and inspected

7.3 Not tested

- **Official W3C validation:** the pages were not submitted to the W3C Markup Validation Service.
- **Colour contrast:** not measured. The pages use the browser's default colours because there is no CSS.
- **Screen-reader behaviour:** not tested with a screen reader.
- **Keyboard use in the browser:** not tested by hand; item 11 is based on reviewing the source.
- **Other browsers and devices:** only Microsoft Edge on Windows 11 was used; phones and other browsers were not tested.

8. Results and Conclusion

Seventeen static HTML5 pages were developed for StudentHub: the eleven pages required by the problem definition (Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ and Feedback) and six additional student-area pages. All pages share one semantic skeleton with a skip-to-content link, a header, a main navigation that marks the current page, a breadcrumb, a single main element and a footer, so the structure is consistent across the portal. Forms use labels, fieldsets and legends, the logo image has descriptive alt text, and every page is reachable from every other page.

The local checker passed all 17 pages, and the browser screenshots of all pages were captured and inspected. The breadcrumb (intermediate extension) and the skip-to-content link with keyboard-friendly markup (advanced extension) from the practical list are included.

Remaining limitations: the forms do not send data because there is no server in this practical; the pages contain sample data only; and official W3C validation, colour contrast, screen-reader testing and manual keyboard testing were not carried out. Styling with CSS Grid, Flexbox and a framework follows in Practical 3.

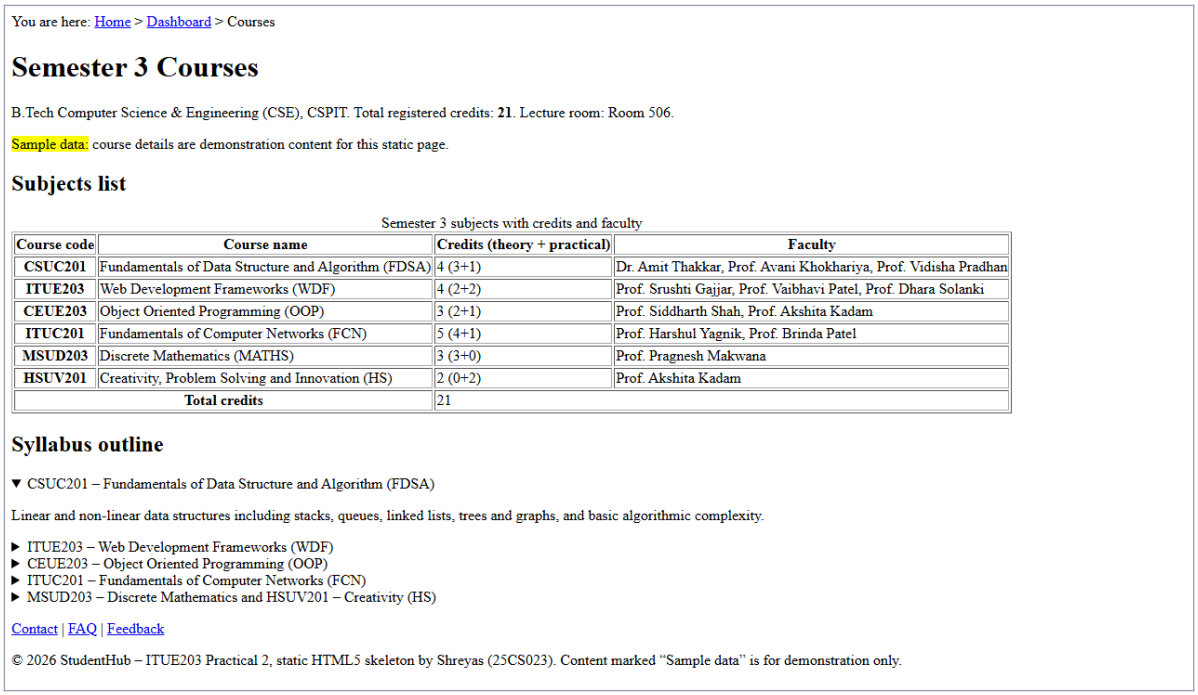
9. References

- ITUE203 Web Development Frameworks – Practical List 2026-27, Practical 2: "Semantic HTML5 Pages with Accessibility-Ready Structure", CSPIT, CHARUSAT.

- StudentHub repository: https://github.com/TheEmperor08/StudentHub-ITUE203-25CS023 (this practical's changes are uploaded separately).
- Practical-2/README.md – page list and accessibility checklist; Practical-2/source/check_pages.py – local checker.
- WHATWG, HTML Living Standard – https://html.spec.whatwg.org/
- W3C Markup Validation Service – https://validator.w3.org/ (recommended for official validation; not used in this report).
- Tools: Microsoft Edge 156.0.4314.50, Python 3.12.0, Visual Studio Code, Git.

Appendix A. Additional Pages

Browser output of the six additional pages. To keep this appendix compact, each image is cropped below the shared header and navigation (identical on every page, see Figure 1), so it starts at the breadcrumb. The full screenshots are 12-courses.png to 17-directory.png in Practical-2/screenshots/browser/.



You are here: Home > Dashboard > Courses

Semester 3 Courses

B.Tech Computer Science & Engineering (CSE), CSPIT. Total registered credits: **21**. Lecture room: Room 506.

Sample data: course details are demonstration content for this static page.

Subjects list

Semester 3 subjects with credits and faculty

Course code	Course name	Credits (theory + practical)	Faculty
CSUC201	Fundamentals of Data Structure and Algorithm (FDSA)	4 (3+1)	Dr. Amit Thakkar, Prof. Avani Khokhariya, Prof. Vidisha Pradhan
ITUE203	Web Development Frameworks (WDF)	4 (2+2)	Prof. Srushti Gajjar, Prof. Vaibhavi Patel, Prof. Dhara Solanki
CEUE203	Object Oriented Programming (OOP)	3 (2+1)	Prof. Siddharth Shah, Prof. Akshita Kadam
ITUC201	Fundamentals of Computer Networks (FCN)	5 (4+1)	Prof. Harshul Yagnik, Prof. Brinda Patel
MSUD203	Discrete Mathematics (MATHS)	3 (3+0)	Prof. Pragnesh Makwana
HSUV201	Creativity, Problem Solving and Innovation (HS)	2 (0+2)	Prof. Akshita Kadam
Total credits		21	

Syllabus outline

▼ CSUC201 – Fundamentals of Data Structure and Algorithm (FDSA)

Linear and non-linear data structures including stacks, queues, linked lists, trees and graphs, and basic algorithmic complexity.

▶ ITUE203 – Web Development Frameworks (WDF)
▶ CEUE203 – Object Oriented Programming (OOP)
▶ ITUC201 – Fundamentals of Computer Networks (FCN)
▶ MSUD203 – Discrete Mathematics and HSUV201 – Creativity (HS)

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Figure A1: Courses page (courses.html), below the shared header

You are here: Home > Dashboard > Attendance

Attendance Record

Shreyas (25CS023) | CSE Department, CSPIT | Semester 3

Sample data: attendance figures on this page are placeholders, not actual records.

Subject-wise attendance

A minimum of 75% attendance is required to appear for end-semester examinations.

Subject-wise attendance, average 94.0%

Subject	Lectures held	Lectures attended	Percentage	Meter
CSUC201 – Data Structure and Algorithm (FDSA)	20	19	95%	
ITUE203 – Web Development Frameworks (WDF)	25	24	96%	
CEUE203 – Object Oriented Programming (OOP)	15	14	93.3%	
ITUC201 – Computer Networks (FCN)	25	23	92%	
MSUD203 – Discrete Mathematics (MATHS)	17	16	94.1%	
HSUV201 – Creativity, Problem Solving (HS)	16	15	93.7%	
Total	118	111	94.0%	

Apply for leave

Leave application

Date of absence

dd-mm-yyyy

Leave type

Hackathon or technical competition (on duty)

Details

Note: This is a static form skeleton. Practical 2 has no server, so the "Submit leave request" button is disabled and nothing is sent or saved.

Submit leave request

Contact | FAQ | Feedback

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Figure A2: Attendance page (attendance.html), below the shared header

You are here: Home > Dashboard > Assignments

Assignments

Shreyas | B.Tech CSE, Semester 3, CSPIT

Sample data: assignment status on this page is demonstration content.

Assignment status

Notice: theory problem sets and laboratory manuals for Semester 3 are being prepared by the course faculty. Assignments are coming soon.

Semester 3 assignment status by subject

Subject	Faculty coordinator	Status	Submission
CSUC201 – Data Structure and Algorithm (FDSA)	Prof. Avani Khokhariya / Dr. Amit Thakkar	Coming soon	Not open yet
ITUE203 – Web Development Frameworks (WDF)	Prof. Srushti Gajjar / Prof. Vaibhavi Patel	Coming soon	Not open yet
CEUE203 – Object Oriented Programming (OOP)	Prof. Siddharth Shah / Prof. Akshita Kadam	Coming soon	Not open yet
ITUC201 – Computer Networks (FCN)	Prof. Harshul Yagnik / Prof. Brinda Patel	Coming soon	Not open yet

Submit an assignment

Once assignments are released, solutions will be uploaded here.

Assignment submission

Subject

-- Choose subject --

Solution file (PDF or ZIP)

Choose File No file chosen

Note: This is a static form skeleton. Practical 2 has no server, so the "Upload submission" button is disabled and nothing is sent or saved.

Upload submission

Contact | FAQ | Feedback

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Figure A3: Assignments page (assignments.html), below the shared header

You are here: Home > Dashboard > Results

Examination Results

Shreyas | Enrollment: 25CS023 | B.Tech CSE, CSPIT

Sample data: grades on this page are placeholders, not actual academic results.

Cumulative performance

Cumulative Grade Point Average (CGPA): **8.05 / 10.0**

Semester-wise SGPA and CGPA

Semester	SGPA	CGPA	Result
Semester 1	8.32	8.32	Passed
Semester 2	7.77	8.05	Passed
Semester 3 (current)	Ongoing	Ongoing	21 credits registered

Semester 2 grade summary

Semester 2 grades by subject category, SGPA 7.77

Subject category	Evaluation type	Grade
Core engineering subjects	Theory and practical exams	AA / AB
Mathematics and foundation	End-semester theory	AB / BB
Laboratory and workshops	Practical evaluation	AA

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Figure A4: Results page (results.html), below the shared header

You are here: Home > Dashboard > Notices

Campus Notices

Sample data: the notices below are example content for this static page.

Latest notices

Odoo Hackathon at LDCE on 22 August

The Odoo Hackathon is scheduled at L.D. College of Engineering on 22 August 2026. CSE students interested in open-source development are encouraged to form teams. Details are on the Events page.

Posted on 5 August 2026 | Category: Hackathon

Semester 3 timetable (with effect from 6 July)

The revised timetable for B.Tech Semester 3 CSE has been released. Most theory lectures are in Room 506 (A7 Building); practical batches A1, B1 and C1 attend laboratories 631, 632 and 633. See the Dashboard.

Posted on 6 July 2026 | Category: Academic

Independence Day celebration

All faculty members and students are invited to celebrate Independence Day on 15 August 2026 at the central university grounds.

Category: Campus event

Contact | FAQ | Feedback

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Figure A5: Notices page (notifications.html), below the shared header

You are here: Home > Dashboard > Directory

Campus Directory

Sample data: all entries are fictional placeholders. The directory contains no real student data.

Search the directory

Name or enrollment number

Course
All courses

Note: This is a static form skeleton. Practical 2 has no server, so the "Search" button is disabled and nothing is sent or saved.

Search

Students

Student directory (sample entries)

Name	Enrollment	Course	Year
Sample Student 1	DEMO-001	B.Tech CSE	2nd year
Sample Student 2	DEMO-002	B.Tech CSE	2nd year
Sample Student 3	DEMO-003	B.Tech IT	2nd year
Sample Student 4	DEMO-004	B.Tech CE	3rd year
Sample Student 5	DEMO-005	BCA	1st year

Department contacts

For department and helpdesk details, see the Contact page.

Contact | FAQ | Feedback

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Figure A6: Directory page (directory.html), below the shared header